

Figure 2. Both hot kernels ran far below the memory roofline.

Each had been written as a matrix-matrix product and was being used at decode as a matrix-vector product: `tl.dot` with a 16-row accumulator to multiply a single token, and packed weights indexed with the output dimension on the fastest-varying axis, so neighbouring lanes in a warp addressed separate cache lines. The dense kernel gains less because a single 5.31 MB matrix yields only 64 thread blocks against 142 streaming multiprocessors, leaving it starved for parallelism rather than limited by memory.

Source: `engine/kernels/RESULTS.md`

